

1(a)	work done per unit mass	B1
	(work done to) move mass from infinity (to the point)	B1
1(b)	curve from r to $4r$, with gradient of decreasing magnitude and starting at $(r, \pm\phi)$	B1
	line passing through $(2r, \pm0.5\phi)$ and $(4r, \pm0.25\phi)$	B1
	line showing potential is negative throughout	B1
1(c)(i)	gravitational potential energy = $(-)\ GMm / R$	C1
	change = $(6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 3.4 \times 10^3) / (6.4 \times 10^6) \times [1/3 - 1/4]$	C1
	= 1.8×10^{10} J	A1
1(c)(ii)	rock loses potential energy	B1
	(so) kinetic energy increases so speed increases	B1
	or	
	force is attractive	(B1)
	moves towards planet so speeds up	(B1)